Class 9 Mathematics — Exercise 4.8

Basic Concept

Formulas Used

Shape / Quantity	Formula
Area of rectangle	$A = l \times w$
Perimeter of rectangle	$P = l + l + w + w = 2(l + w)$
Area of square	$A = l \times l = l^2$
Perimeter (boundary) of square	$P = 4l$
Volume of cube	$V = l \times l \times l = l^3$
Surface area of cube	$6(l \times l) = 6l^2$
Area of circle	$A = \pi r^2$
Distance	$d = v \times t$
Cost	Rate $\times$ Quantity

Method: Factorize the given expression; the factors give the unknown dimensions.

Question 1

In a map of an industry, the area of a rectangular veranda is given by $(x^2 - 2x - 3)\,\mathrm{m}^2$. Find: (a) expression for both dimensions of the veranda (b) expression for perimeter of the veranda (c) expression for cost of fencing the veranda @ Rs. 200/m (d) expression for cost of carpeting the veranda floor @ Rs. 250/m².

$$\text{Area} = (x^2 - 2x - 3)\,\mathrm{m}^2, \qquad A = l \times w$$

(a) $-3 \to -3$ and $1 \quad (-3 + 1 = -2)$

$$x^2 - 2x - 3 = x^2 - 3x + x - 3 = x(x - 3) + 1(x - 3) = (x - 3)(x + 1)$$

$$\boxed{l = (x - 3)\,\mathrm{m}, \qquad w = (x + 1)\,\mathrm{m}}$$

(b)

$$P = 2(l + w) = 2[(x - 3) + (x + 1)] = 2(2x - 2) = \boxed{(4x - 4)\,\mathrm{m}}$$

(c) Cost of fencing @ Rs. 200/m:

$$= 200 \times (4x - 4) = \boxed{\text{Rs. } (800x - 800)}$$

(d) Cost of carpeting @ Rs. 250/m²:

$$= 250 \times (x^2 - 2x - 3) = \boxed{\text{Rs. } 250(x^2 - 2x - 3)}$$

Question 2

Area of a square shaped surface of a machine is given by $(25x^2 - 30x + 9)\,\text{m}^2$. Find: (a) expression for the length of the surface (b) expression for the boundary of the surface (c) expression for the cost of polishing the surface @ Rs. 75/m² (d) expression for the cost of edging around 2 sides of the machine surface @ Rs. 28/m.

$$\text{Area} = (25x^2 - 30x + 9)\,\text{m}^2, \qquad A = l^2$$

(a) ($\because a^2 - 2ab + b^2 = (a - b)^2$)

$$25x^2 - 30x + 9 = (5x)^2 - 2(5x)(3) + (3)^2 = (5x - 3)^2$$

$$l^2 = (5x - 3)^2 \;\Rightarrow\; \boxed{l = (5x - 3)\,\text{m}}$$

(b) Boundary $= 4l = 4(5x - 3) = \boxed{(20x - 12)\,\text{m}}$

(c) Cost of polishing @ Rs. 75/m²:

$$= 75(25x^2 - 30x + 9) = \boxed{\text{Rs. } 75(5x - 3)^2}$$

(d) Length of 2 sides $= 2(5x - 3)$. Cost of edging @ Rs. 28/m:

$$= 28 \times 2(5x - 3) = 56(5x - 3) = \boxed{\text{Rs. } (280x - 168)}$$

Question 3

Volume of a cubical oil tank in an oil refinery is $(125x^3 - 150x^2 + 60x - 8)\,\text{m}^3$. Find: (a) expression for height of oil tank (b) expression for surface area of oil tank (c) expression for painting it from outside @ Rs. 32/m².

$$\text{Volume} = (125x^3 - 150x^2 + 60x - 8)\,\text{m}^3, \qquad V = l^3$$

(a) ($\because a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3$)

$$125x^3 - 150x^2 + 60x - 8 = (5x)^3 - 3(5x)^2(2) + 3(5x)(2)^2 - (2)^3 = (5x - 2)^3$$

$$l^3 = (5x - 2)^3$$

Taking cube root on both sides:

$$l = \sqrt[3]{(5x-2)^3} = (5x-2)\,\text{m}$$

$$\boxed{\text{height} = (5x-2)\,\text{m}}$$

(b) Surface area of oil tank $= 6(l \times l)$

$$= 6(5x-2)(5x-2) = \boxed{6(5x-2)^2\,\text{m}^2}$$

(c) Cost of painting @ Rs. 32/m^2:

$$= 32 \times 6(5x-2)^2 = \boxed{\text{Rs. } 192(5x-2)^2}$$

Question 4

A mechanical engineer working on wheels of a machine finds their areas given by $A_1 = \pi x^2 - 6\pi x + 9\pi$ and $A_2 = \pi x^2 - 10\pi x + 25\pi$. Help him find the radii of both wheels.

Area of circle $= \pi r^2$

$$A_1 = \pi x^2 - 6\pi x + 9\pi = \pi(x^2 - 6x + 9) = \pi[(x)^2 - 2(x)(3) + (3)^2] = \pi(x-3)^2$$

$$\pi r_1^2 = \pi(x-3)^2 \;\Rightarrow\; \boxed{r_1 = x - 3}$$

$$A_2 = \pi x^2 - 10\pi x + 25\pi = \pi(x^2 - 10x + 25) = \pi[(x)^2 - 2(x)(5) + (5)^2] = \pi(x-5)^2$$

$$\pi r_2^2 = \pi(x-5)^2 \;\Rightarrow\; \boxed{r_2 = x - 5}$$

Question 5

A machine has two square shaped pressers with areas $25m^2$ and $36n^2$ respectively. The difference of these areas describes a rectangular presser. Find the dimensions of the rectangular presser.

$$A_1 = 25m^2, \qquad A_2 = 36n^2$$

Area of rectangular presser:

$$A = A_1 - A_2 = 25m^2 - 36n^2$$
$$= (5m)^2 - (6n)^2 \qquad (\because\ a^2 - b^2 = (a+b)(a-b))$$
$$= (5m + 6n)(5m - 6n)$$

Since $A = l \times w$:

$$\boxed{\text{Dimensions} = (5m + 6n) \text{ and } (5m - 6n)}$$

Question 6

Distance covered by a missile to hit the target is given by $(x^2 + 5x + 6)$ m. Find: (a) the possible expression for speed of the missile (b) the possible expression for time to reach the target.

$$\text{Distance} = (x^2 + 5x + 6)\,\text{m}, \qquad d = v \times t$$

$6 \to 3$ and $2 \quad (3 + 2 = 5)$

$$\begin{aligned} d &= x^2 + 3x + 2x + 6 \\ &= x(x + 3) + 2(x + 3) \\ d &= (x + 3)(x + 2) \end{aligned}$$

(a), (b)

If speed $= x + 2$ then time $= x + 3$, and vice versa